# StockTrader: Comparative Analysis Report

*Momentum Trader vs. Value Contrarian — Multi-Agent Stock Analysis*

*Course bonus: third strategy — Volatility Averse — see §8*

## 1. Strategy Selection and Rationale

The **required** assignment compares two strategies with fundamentally incompatible philosophies: the **Momentum Trader** (trend-following) and the **Value Contrarian** (mean-reversion).

The Momentum Trader believes that “the trend is your friend.” It prioritizes price position relative to 20-day and 50-day moving averages, volume trend confirmation, positive-to-negative day ratios, and RSI as a momentum gauge. It explicitly ignores 52-week distances, drawdowns, and volatility. Its personality is aggressive and action-oriented, biased toward BUY and SELL over HOLD.

The Value Contrarian believes that “markets overreact — buy fear, sell greed.” It prioritizes distance from 52-week extremes, RSI as an overbought/oversold indicator, drawdown magnitude, and volatility as a signal of emotional trading. It explicitly ignores moving average crossovers, volume trends, and short-term win/loss ratios. Its personality is patient and skeptical, biased toward HOLD over action.

For the **optional bonus (third strategy agent)**, I added **Volatility Averse** from the course Strategy Menu: it prioritizes **path risk** — standard deviation of daily returns (30d/90d), ATR as a percent of price, worst single-day drop over 90 days, and max drawdown — and de-emphasizes trend vs. contrarian framing. It runs in parallel with the other two on the same market data.

This pairing was chosen because the two strategies apply opposing interpretive frameworks to identical data. A stock trading above both moving averages with strong volume (Momentum’s ideal BUY signal) could simultaneously be near its 52-week high with an elevated RSI (Contrarian’s ideal SELL signal). The expected disagreement is structural, not random — it emerges from the incompatibility of their data hierarchies. I expected sharp disagreement on momentum-driven stocks (NVDA, TSLA) and potential convergence on neutral stocks (JNJ, GME). I added **GME** (meme/retail volatility) and **JPM** (large bank, rate-credit narrative) to stress-test the prompts on non-tech regimes.

## 2. System Architecture

The system uses a parallel fan-out pipeline with four components orchestrated in plain Python with asyncio:

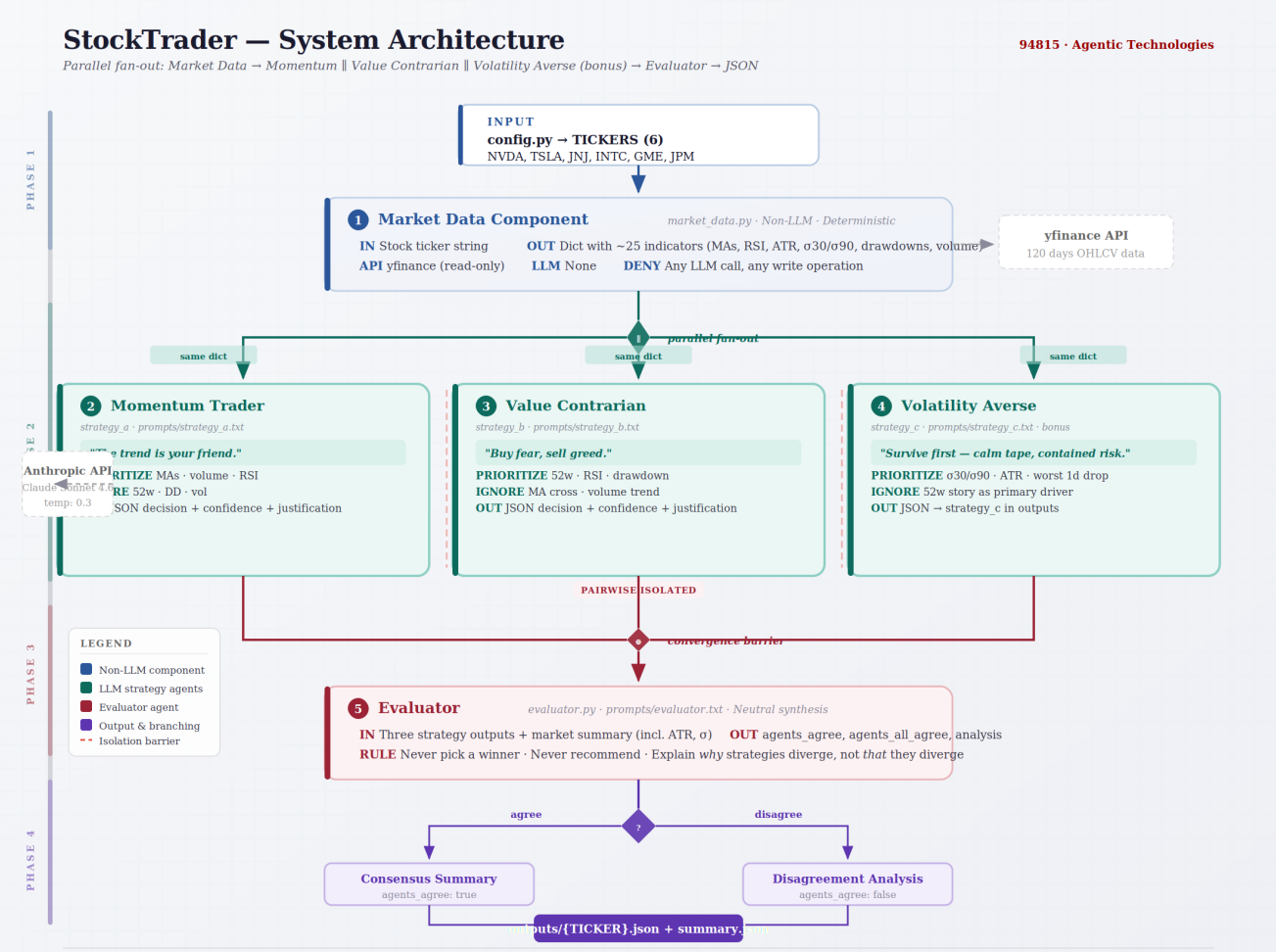



*Figure 1: StockTrader system architecture — parallel fan-out with convergence barrier.*

**Phase 1 — Data:** The Market Data Component (non-LLM, deterministic) fetches 120 days of OHLCV data from yfinance and computes MAs, RSI, 30d/90d volatility, 14-day ATR, worst single-day drop, drawdowns, and volume trends. Zero LLM calls. **Phase 2 — Analysis:** Three strategy agents receive the identical data dict via `asyncio.gather`. No agent sees another's output. All use Claude Sonnet (`claude-sonnet-4-6`) at temperature 0.3. **Phase 3 — Evaluation:** Agreement is computed programmatically; the Evaluator produces a three-way synthesis without picking a winner. **Phase 4 — Output:** Structured JSON per stock plus `summary.json`. No agent framework — plain Python orchestration.

## 3. Stock Selection and Rationale

Six stocks were selected to represent distinct market conditions:

| Ticker | Market Condition | Expected Behavior |
|---|---|---|
| NVDA | Strong uptrend / momentum | Momentum BUY vs. Contrarian caution near highs. Sharp disagreement expected. |
| TSLA | Volatile, large swings | Mixed signals; disagreement likely but direction unpredictable. |
| JNJ | Stable large-cap, low vol | Possible convergence on HOLD — no strong trend or extreme positioning. |

| Ticker | Market Condition | Expected Behavior |
|---|---|---|
| INTC | Selected for decline; on run date captured parabolic rally to ATH | Momentum conflicted HOLD (RSI override). Contrarian + Vol Averse SELL. Different axis than expected. |
| GME | Meme / retail volatility | Extreme swings; tests whether justifications stay data-grounded. |
| JPM | Large U.S. bank | Cyclical, rate-sensitive; contrasts with tech-heavy set. |

### 3.1 Snapshot aligned to this run (2026-04-12)

- **NVDA:** Three-way split (BUY / HOLD / SELL). Vol Averse rejects tape as too turbulent (ATR ~2.75% of price).
- **TSLA:** Momentum + Vol Averse SELL; Contrarian BUY. Classic trend vs. fear.
- **JNJ:** Momentum HOLD, Contrarian HOLD, Vol Averse BUY (low realized vol).
- **INTC:** Parabolic/ATH. Contrarian + Vol Averse SELL; Momentum conflicted HOLD.
- **GME:** Unanimous HOLD — only full three-way agreement in the run.
- **JPM:** Contrarian SELL; Momentum HOLD, Vol Averse HOLD (2–1 split).

## 4. Results by Stock

**Run date:** 2026-04-12. Complete JSON lives in `stocktrader/outputs/`. This section provides a summary table, short excerpts for four tickers, and two verbatim JSON excerpts (TSLA, JNJ). Full JSON for all six tickers appears in Appendix B.

### 4.1 Summary — all six stocks

| Ticker | Momentum | Contrarian | Vol. Averse | A/B agree? | All 3? |
|---|---|---|---|---|---|
| NVDA | BUY | HOLD | SELL | No | No |
| TSLA | SELL | BUY | SELL | No | No |
| JNJ | HOLD | HOLD | BUY | Yes | No |
| INTC | HOLD | SELL | SELL | No | No |
| GME | HOLD | HOLD | HOLD | Yes | Yes |
| JPM | HOLD | SELL | HOLD | No | No |

**Momentum vs Contrarian:** 2 agreements / 4 disagreements. **Unanimous (all three):** 1 (GME only). The only opposite-action pair (SELL vs BUY) is TSLA.

### 4.2 Short excerpts (NVDA, INTC, GME, JPM)

**NVDA (three-way split).** Vol Averse: “NVDA’s 30-day volatility of 2.242% and 90-day volatility of 2.1607% both sit at elevated levels… The ATR as a percentage of price at 2.7541% means daily swings are averaging nearly 3% of the stock’s value.”

**INTC (HOLD / SELL / SELL).** Momentum Trader: “INTC is in a screaming uptrend — up 53.8% over 90 days and sitting at its 52-week high… but the RSI of 80.35 is a serious red flag that screams overbought exhaustion, well above my 75 threshold.”

**GME (unanimous HOLD).** All three cite lack of a clean trigger; Vol Averse emphasizes ATR % and drawdown.

**JPM (2–1).** Vol Averse HOLD with Momentum vs Contrarian SELL; path metrics (ATR, 90d vs 30d vol) justify pausing.

### 4.3 Direct excerpt 1 — TSLA (strategies + evaluator)

Momentum Trader: **SELL** (confidence 8/10). “TSLA is trading at $348.95, sitting -5.9% below its 20-day moving average and -11.5% below its 50-day moving average — a clear bearish positioning that signals the trend has reversed. The RSI of 41.97 is approaching oversold territory but hasn’t bounced, confirming sustained selling pressure.”

Value Contrarian: **BUY** (confidence 6/10). “TSLA is sitting 28.77% below its 52-week high of $489.88, firmly in the territory where fear-driven selling creates contrarian opportunity. The 90-day max drawdown of -29.93% suggests meaningful capitulation has occurred.”

Volatility Averse: **SELL** (confidence 8/10). “TSLA’s 30-day volatility of 2.38% and 90-day volatility of 2.32% are both elevated… The ATR as a percentage of price stands at 4.42%… a chaotic, high-risk ride.”

Evaluator: “The split reflects a fundamental philosophical divide: Contrarian weights distance from peak and potential capitulation, while Momentum and Volatility Averse both weight current conditions — one through directional signal, the other through path risk — and both conclude the environment demands an exit.”

### 4.4 Direct excerpt 2 — JNJ (strategies + evaluator)

Momentum Trader: **HOLD** (confidence 4/10). “JNJ is trading at $238.46, sitting just below both the 20-day MA ($239.90) and 50-day MA ($240.24)… Volume is decreasing and negative days (16) outnumber positive days (14).”

Value Contrarian: **HOLD** (confidence 7/10). “JNJ is only 4.06% below its 52-week high, far from the 20–30% drawdown threshold… RSI of 56.86 sits comfortably in the neutral zone.”

Volatility Averse: **BUY** (confidence 7/10). “JNJ’s 30-day volatility of 0.9844% and 90-day volatility of 1.0479% are both modest… ATR at just 1.7703% of price… a controlled, low-turbulence name.”

Evaluator: “The outlier’s logic is path-focused rather than directional — it does not require trend strength or fear pricing, only the absence of dangerous volatility. This divergence illustrates how the same data point — modest recent price decline — is read as a warning flag by one lens, an irrelevant non-event by another, and a non-concern by the third.”

## 5. Patterns of Agreement and Disagreement

### Where Strategies Agreed

**Momentum vs Contrarian** agreed on **JNJ** and **GME** (both HOLD). **All three** agreed only on **GME** (unanimous HOLD). JNJ’s agreement masked a deeper split: Vol Averse issued BUY based on the smooth tape (30d vol 0.98%, ATR 1.77% of price), while the other two found nothing to act on.

**GME** ($23.22) was the run’s only full unanimity. Each lens found a different reason not to act: mixed MAs, no capitulation, ATR/drawdown still uncomfortable. The sideways drift — essentially flat over 90 days at +3.06% — simultaneously failed the entry criteria of all three disciplines.

### Where Strategies Disagreed Most Sharply

The clearest Momentum-vs-Contrarian fight is **TSLA** (SELL vs BUY). Volatility Averse also SELLs, aligning with Momentum on direction while citing ATR %, daily volatility, and worst single-day drop

— not trend labels.

**NVDA** is the run's three-way split (BUY / HOLD / SELL): trend vs "too rich" vs "tape too turbulent." **INTC** shows Contrarian + Vol Averse aligned on SELL (euphoria vs chaotic path) while Momentum HOLDs. **JPM** is a 2–1: Contrarian SELL vs HOLD/HOLD.

### Cross-Stock Patterns (own analysis)

Two patterns emerged that the evaluator did not articulate. First, **RSI is the systematic divergence trigger** across the entire run. Every Momentum-vs-Contrarian disagreement involves an RSI reading that one agent interprets as momentum confirmation and the other as an overbought/oversold extreme: NVDA (68.17 — "sweet spot" vs. "approaching overbought"), TSLA (41.97 — "sustained selling pressure" vs. "approaching oversold opportunity"), INTC (80.35 — "overbought exhaustion" vs. "peak euphoria to sell"), JPM (73.69 — "nearing exhaustion" vs. "firmly overbought"). RSI is the single indicator that both strategies include but interpret through opposite lenses — making it the fulcrum of disagreement by design.

Second, the **Momentum Trader was far more conservative than expected**. It issued HOLD on 4 of 6 stocks, never once producing its "preferred" aggressive BUY on a clear trend stock. Even INTC — at its 52-week high, 20/30 positive days, increasing volume, price 29.5% above the 20-day MA — only earned a conflicted HOLD at 5/10 confidence because RSI at 80.35 tripped the overbought safety valve. The prompt's RSI ceiling (75) effectively overrode every other momentum signal in the dataset, biasing the agent toward caution in a way that makes it behaviorally closer to the Contrarian's patience than intended. A future iteration might relax the RSI ceiling or make it a "flag" rather than a hard gate.

This matches the assignment's point about **behavioral design**, not forecasting skill: agents disagree because they are instructed to weight different columns of the same table, and even internal prompt constraints (like RSI thresholds) shape behavior as much as the philosophical framing does.

## 6. Failure or Surprise Case

**Primary case: Momentum Trader's RSI safety valve overrides its strongest BUY signal.**

**(1) What happened.** On **INTC**, the Momentum Trader's own framework produced the strongest BUY setup in the entire dataset: 52-week high ($62.38), price 29.5% above the 20-day MA, 20/30 positive days, increasing volume, +53.8% over 90 days. Every momentum signal pointed to BUY. Yet it issued **HOLD at 5/10 confidence** — its lowest-conviction call — because RSI at 80.35 tripped the 75 threshold in the prompt.

**(2) Why it happened.** The prompt treats RSI > 75 as a hard gate rather than a weighted input. When RSI breaches 75, it vetoes every other indicator regardless of how bullish they are. This is a **prompt design tension**: the RSI ceiling makes the agent safer but also makes it unable to act on parabolic trends — exactly the moves a momentum philosophy should chase.

**(3) What I would change.** Restructure so RSI > 75 triggers a **confidence discount** (cap at 5–6) rather than a decision override. This preserves the behavioral identity while still flagging exhaustion risk.

**Secondary: Evaluator language drift.** On INTC and JPM, the evaluator describes the Value Contrarian as "more internally consistent," violating its neutral role. Fix: add hard negatives to the prompt.

**Tertiary: Persona leakage.** On JPM, the Contrarian cites distance from the 20-day MA — an indicator its prompt should de-emphasize. A small prompt-compliance slip for a future revision.

## 7. Reflection

**If I had $10,000 and had to follow one strategy for the next month:** I would choose the **Value Contrarian** because its recommendations tracked the mean-reversion story most transparently: it sold into euphoria on INTC (52-week high, RSI ~80) and JPM (RSI ~73.7, near highs), and bought TSLA after a 28.77% drawdown from the 52-week high. The Momentum Trader was more conservative than expected — HOLD on 4/6 stocks — making it less suitable for a single decisive month but more suitable for avoiding parabolic traps.

**A third hybrid strategy** would use the Momentum Trader's framework as the primary decision-maker but apply the Contrarian's RSI and 52-week checks as a risk filter: override BUY when RSI > 75 or within 5% of the 52-week high; override SELL when RSI < 30 or >25% below the 52-week high. This captures trend-following while avoiding buying at tops and selling at bottoms.

## 8. Bonus — Third Strategy Agent (Volatility Averse)

For the course bonus, I implemented the **Third Strategy Agent** using **Volatility Averse** from the official Strategy Menu (key inputs: std dev of daily returns, ATR, worst single-day drop — all from yfinance OHLCV). It runs in parallel on the same data; outputs add `strategy_c` and `three_way_pattern` to each JSON, and `summary.json` records `total_all_three_agree` (1 of 6: GME only).

What this added: **three-way splits** (NVDA: BUY/HOLD/SELL) show that path-risk framing is not reducible to the two-way story. TSLA: Momentum and Vol Averse both SELL, but for different reasons (trend vs ride). JNJ: Momentum and Contrarian HOLD while Vol Averse BUYs the smooth tape. The bonus makes disagreement structure richer — unanimous vs 2–1 vs three-way — and surfaces *why* each philosophy fires.

*Appendices A and B are supplementary reference material beyond the 3–5 page report body (Sections 1–8 above).*

# Appendix A: AI Use Documentation

## Tools Used

• **yfinance:** Historical OHLCV for the Market Data component (no LLM calls).

• **Anthropic Messages API (Claude Sonnet,** `claude-sonnet-4-6`**):** Four LLM calls per stock: Momentum Trader, Value Contrarian, Volatility Averse (bonus), Evaluator. Temperature 0.3, single-shot.

• **Claude (Anthropic) in Cursor:** System design, prompt drafting, architecture planning, report structure. All outputs reviewed and edited by the student.

## Representative Prompts

Each strategy agent received a system prompt encoding: behavioral philosophy, explicit decision framework, ranked data priority hierarchy, explicit ignore list, and personality bias. User prompts formatted computed indicators as labeled key-value pairs and demanded JSON-only responses with ≥3 specific data references. Full prompts saved as `prompts/strategy_a.txt`, `strategy_b.txt`, `strategy_c.txt`, `evaluator.txt`.

## What I Accepted, Revised, and Rejected

**Accepted:** Parallel fan-out architecture, Anthropic's Claude Sonnet for inference, stock selection rationale, JSON output structure.

**Revised:** Updated report after inspecting JSON so Section 3's INTC description matches the actual parabolic/ATH snapshot. Expanded to six tickers (added GME, JPM). Added bonus Volatility Averse agent with ATR/volatility fields and three-way evaluator. Documented evaluator and persona issues in Section 6 rather than fixing outputs.

**Rejected:** Did not use multi-agent frameworks (LangChain, CrewAI). Did not alter or discard JSON outputs. Did not treat evaluator's normative phrasing as ground truth — recorded as a limitation.

# Appendix B: Full JSON Output Reference

Complete JSON files for all six tickers are in `stocktrader/outputs/`. Verbatim excerpts for TSLA and JNJ appear in Section 4.3–4.4. Below are the decision summaries and key market data for the remaining four tickers. Full justifications and evaluator analyses are in the repository JSON files.

| Ticker | Price | RSI | 30d Δ% | 52w High dist | Mom. | Contr. | Vol.Av. |
|---|---|---|---|---|---|---|---|
| NVDA | $188.63 | 68.17 | +2.03% | -8.88% | BUY (7) | HOLD (6) | SELL (7) |
| INTC | $62.38 | 80.35 | +37.22% | 0.00% (ATH) | HOLD (5) | SELL (9) | SELL (9) |
| GME | $23.22 | 56.14 | -3.65% | -33.68% | HOLD (4) | HOLD (5) | HOLD (5) |
| JPM | $309.87 | 73.69 | +1.74% | -6.92% | HOLD (5) | SELL (7) | HOLD (6) |